

A-LIS Off-Site Database Backup & Restoration SOP - Linux/Ubuntu

Facility Setup Guide

Version: 2.0

Audience: ICT Officers / A-LIS Implementers

Estimated Setup Time: 15-20 Minutes

Objective

This guide describes how to configure an A-LIS installation to automatically upload compressed database backups to the UNHLS Central Backup Server, and how to restore from those backups if a facility's database instance is lost or corrupted.

Once the backup setup is completed:

- Database backups will run automatically.
- Backups will be copied off-site every day.
- Local backups older than three days will be removed automatically.
- No further user intervention is required.

Prerequisites

Before starting, ensure you have:

- SSH access to the A-LIS server.
- MySQL database credentials.
- Internet/VPN connectivity to the Central Backup Server.
- Backup server IP address.
- Backup server SSH port.
- Shared backup username.

Note: The remote/CPHL backup server IP address and username is made available outside of this guide for security purposes.

“ *Once you have all these in place, scroll to the bottom for a quick Motivational Quote* ”

Step-by-Step Backup Process

1. Create Local Backup Directory & Generate a Dedicated SSH Key

on Facility's ALIS server

Login to the Facility's A-LIS server and Create a backup directory.

i. Create back up directory

```
mkdir ~/.backup
```

ii. While still on the A-LIS facility server generate a dedicated Ed25519 SSH key for the A-LIS off-site backup process:

```
ssh-keygen \  
-t ed25519 \  
-f ~/.ssh/alis_backup_ed25519 \  
-C "alis-offsite-backup"
```

iii. When prompted:

Enter passphrase (empty for no passphrase):

Leave the passphrase empty and press **ENTER**.

When prompted again:

Enter same passphrase again:

Press **ENTER**.

This allows the scheduled backup process to authenticate without requiring manual password entry.

iv. Protect the private SSH key:

```
chmod 600 ~/.ssh/alis_backup_ed25519
```

vi. Verify that the keys were created:

```
ls -lah ~/.ssh/alis_backup_ed25519*
```

The following files should exist:

```
~/.ssh/alis_backup_ed25519  
~/.ssh/alis_backup_ed25519.pub
```

The file **alis_backup_ed25519** is the **private key** and **alis_backup_ed25519.pub** is the **public key** and is the key that must be registered in the CIS Portal.

v. Display the public key:

```
cat ~/.ssh/alis_backup_ed25519.pub
```

Example output:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIxxxxxxxxxxxxxxxxxxxxxxxx alis-offsite-backup
```

Copy the **entire line** and proceed to step 2.

2. Register the Public Key & Download Backup Script From The CIS Portal

Log into the CIS Portal <https://support.cphl.go.ug/>

1. Click Staff Login
2. Enter your email and password to log in
3. On the right hand pane, under **Manager Lists** , select **Infrastructure: A-LIS Remote Backup Keys**

4.

HOME

Welcome, CIS Administrator

Logout

Overview

Support Team Performance

Manager Dashboard

3Assigned Tickets

0In Progress

0Resolved

2SLA Compliant

Recent Tickets

View All

TICKET	SYSTEM	STATUS	DATE LOGGED	ACTIONS
CIS-260706-003	ALIS	Closed	2026-07-06	View
CIS-260706-002	RDS	New	2026-07-06	View
CIS-260706-001	RDS	Closed	2026-07-06	View

Administration

Manage ICT Users

Manager Lists

Infrastructure: A-LIS Remote Backup Keys

A-LIS Key Update Reasons

Systems

Modules

Regions

Facilities

Departments

Designations

Issue Types

Priority/Impact Levels

Ticket Statuses

Resolution Categories

Closure Reasons

INFRASTRUCTURE

A-LIS Remote Backup Keys

Backup configuration re-enabled and provisioned successfully.

Backup Configurations

[Add Backup Configuration](#)

0

Pending

1

Provisioned

0

Failed

Search Backup Configurations

FACILITY	ADDED BY	DATE ADDED	BACKUP DIRECTORY	STATUS	ACTION
Bugambo H/C II Luuka East Central BUGAMBOHCII	CIS Administrator	2026-07-09 11:26	bugambo_hcii	● PROVISIONED	Download Update Deactivate
Kaweweta H/C III Nakaseke District North Buganda KAWEWETA03	CIS Administrator	2026-07-09 14:33	kaweweta_hciiii	● DISABLED	Update Activate

Click **Add Backup Configuration** and fill in the form that Opens. Copy & paste the generated Key in **Step1 vi** above and save.

CPHL ICT SUPPORT PORTAL

INFRASTRUCTURE

A-LIS Remote Backup Configuration

Add Backup Configuration

Save the facility backup settings, provision the central directory, and generate the facility backup script.

Region *
Select region

District *
Select district

Facility *
Select facility

Backup Directory Name *
Generated automatically from the selected facility and validated again on save.

A-LIS Database Name *
alis_db

A-LIS Database Username *
alis_user

A-LIS Database Password *
Enter database password

SSH Public Key *
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA... alis-offsite-backup

Cancel Save Configuration

On successful saving, the facility key status should change to: **Provisioned**, if it fails, status will show **Provisioning failed**

Click the **Download** button on the corresponding Facility Key to download the **backupscrip.sh** file which you will copy to the `~/backup` directory you created in step one.

FACILITY	ADDED BY	DATE ADDED	BACKUP DIRECTORY	STATUS	ACTION
Bugambo H/C II Luuka East Central BUGAMBOHCII	CIS Administrator	2026-07-09 11:26	bugambo_hcii	● PROVISIONED	Download Update Deactivate
Kaweweta H/C III Nakaseke District North Buganda KAWEWETA03	CIS Administrator	2026-07-09 14:33	kaweweta_hciii	● DISABLED	Update Activate

Click the Download button on the corresponding Facility Key row to download the backupscrip.sh file.

Move the downloaded script into the backup directory created in Step 1:

```
mv ~/Downloads/backupscrip.sh ~/backup/
```

Make the script executable:

```
sudo chmod +x ~/backup/backupscrip.sh
```

Change permissions on the ~/.backup directory:

```
sudo chmod -R 755 ~/.backup/
```

Note: These two chmod commands only work once backupscript.sh actually exists on disk, run them after downloading the script from the portal, not before.

3. Test the Backup Set up

i. In the terminal, run:

```
~/.backup/backupscript.sh
```

ii. You should see:

```
Connected to 192.168.1.1 (example IP address, you will see the Remote Server IP here)
Uploading...
```

4. Automate Back up process

Schedule the Backup with Cron: If you want this backup to happen automatically at regular intervals, you can add the script to `cron` to schedule backups.

i. Open cron

```
crontab -e
```

ii. Add a cron job to run the script daily at midnight (adjust as needed):

```
#makes database dump everyday at midnight
0 0 * * * ~/.backup/backupscript.sh >> ~/backup_cron.log 2>&1
```

Verify Backup

i. Log into the remote/CPHL server:

```
sftp cphl_backupuser_name@remote_server_ip_address
```

ii. List/ inspect the directory for your facility's backup

```
ls -la
```

Example output: **alis_2026-07-08_020000.sql.gz**

Motivational Quote:

**“ If it does not work on your computer, tell them it worked on mine.
Or, rather just cry**

Restoring from a Backup

Use this section when a facility's A-LIS database instance is broken, corrupted, or lost, and needs to be restored from the off-site gzip backups produced by the process above.

1. Retrieve the Backup File

Connect to the central backup server and pull down the facility's most recent backup (or an earlier one, if the latest is suspected to be affected):

```
sftp -i ~/.ssh/alis_backup_ed25519 -P 22 backupuser@105.27.247.146

cd /dumps/<facility>

ls -la

get alis_db_name_2026-07-08_02-00-00.sql.gz

bye
```

Note: Replace <facility> and the filename with the actual facility folder and backup you need.

2. Verify the Backup File Integrity

Before touching the live database, confirm the archive isn't corrupted:

```
gzip -t alis_db_name_2026-07-08_02-00-00.sql.gz && echo OK || echo CORRUPT
```

If the check reports CORRUPT, retrieve an earlier dated backup from the same facility folder and repeat this check.

3. Safety-Dump the Existing Database (if it's still reachable)

If the current instance is still partially running, take one more dump before overwriting anything, in case the "broken" instance still holds data worth keeping:

```
mysqldump -u alis_db_username -p alis_db_name > pre_restore_safety_dump.sql \
```

```
2>/dev/null || echo "skip — instance unreachable"
```

4. Recreate the Database (if needed)

If the database no longer exists on the target MySQL instance:

```
mysql -u root -p -e "CREATE DATABASE IF NOT EXISTS alis_db_name;"
```

5. Restore the Backup

```
gunzip -c alis_db_name_2026-07-08_02-00-00.sql.gz | mysql -u alis_db_username -p alis_db_name
```

This decompresses the backup and pipes the SQL directly into MySQL. Depending on database size, this can take a few minutes.

6. Confirm the Restore

- Log in and spot-check row counts on a couple of key tables.
- Open the A-LIS application itself and confirm it connects and loads recent records.
- Check the application logs for connection or schema errors.

Restoration Notes

- If MySQL will not start at all (not just a corrupted database), fix or reinstall MySQL first, confirm `mysql -u root -p` works, then proceed with the restore.
- If restoring to a different server than the original, check MySQL version compatibility — routines and triggers from a newer MySQL version can fail to import on an older one.
- Test this restore procedure once on a non-production machine so the steps are familiar before they're ever needed under pressure.

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